

SIMPLE COMMANDS:

TOPIC 2 : Directory Management

1. ls

```
Last login: Wed Aug 12 17:52:31 on ttys007
root@Vishnu:~# ls
abc      bookflow_api  Documents     Music         Public
abc.c    bookflow_fastapi  Downloads     mysql         PyCharmMiscProject
abc.c#y  CREATE       dsa 3 odd sem  OSSP          PycharmProjects
Applications  data        Library       package-lock.json  show
blockhack hackathon Desktop      Movies        Pictures
root@Vishnu:~#
```

2. ls-l

```
root@Vishnu:~# ls -l
total 144
-rwxr-xr-x  1 amrutha  staff  33424 Jul 31 13:54 abc
-rw-r--r--  1 amrutha  staff   80 Jul 31 13:51 abc.c
-rw-r--r--  1 amrutha  staff   81 Jul 29 12:39 abc.c#y
drwx-----@ 5 amrutha  staff  160 Jun 19 10:23 Applications
drwxr-xr-x  2 amrutha  staff   64 Aug  7 09:39 blockhack hackathon
drwxr-xr-x  3 amrutha  staff   96 Aug 12 15:31 bookflow_api
drwxr-xr-x  5 amrutha  staff  160 Aug 12 17:20 bookflow_fastapi
-rw-r--r--  1 amrutha  staff 17827 Jul 30 11:02 CREATE
drwxr-xr-x  3 amrutha  staff   96 Aug  3 14:53 data
drwx-----@ 138 amrutha  staff  4416 Aug 12 17:59 Desktop
drwx-----@  5 amrutha  staff  160 Aug 11 13:52 Documents
drwx-----@ 337 amrutha  staff 10784 Aug 12 18:09 Downloads
drwxr-xr-x  3 amrutha  staff   96 Jul 23 12:20 dsa 3 odd sem
drwx-----@ 91 amrutha  staff  2912 Jul 25 08:50 Library
drwx-----  5 amrutha  staff  160 Apr 22 21:03 Movies
drwx-----+  4 amrutha  staff  128 Mar 20 10:47 Music
-rw-r--r--  1 amrutha  staff   51 Jul 30 11:02 mysql
drwxr-xr-x 17 amrutha  staff  544 Aug  3 12:41 OSSP
-rw-r--r--@  1 amrutha  staff   86 May 11 10:41 package-lock.json
drwx-----+  6 amrutha  staff  192 Jun 13 20:31 Pictures
drwxr-xr-x+  4 amrutha  staff  128 Feb 21 12:34 Public
drwxr-xr-x@  4 amrutha  staff  128 Apr 16 14:18 PyCharmMiscProject
drwxr-xr-x@  3 amrutha  staff   96 Apr 16 14:18 PycharmProjects
-rw-r--r--  1 amrutha  staff    0 Jul 30 11:03 show
root@Vishnu:~#
```

3. ls -a

```
root@Vishnu:~# ~ % ls -a
.          .vs_store          .npmrc          abc.c          Movies
..         .gemini           .oracle_jre_usage abc.c#y        Music
.antigravity .gitconfig        .psql_history   Applications   mysql
.antigravity-ide .homebrew         .redhat         blockhack hackathon OSSP
.anydesk      .idlerc           .ssh            bookflow_api   package-lock.json
.arduinoIDE   .ipython          .Trash          bookflow_fastapi Pictures
.bash_history .junie            .vscode         CREATE         Public
.cache        .library          .vscode-shared  data          PyCharmMiscProject
.CFUserTextEncoding .lima            .zprofile       Desktop       PyCharmProjects
.claude       .local            .zprofile.pysave Documents      show
.codex        .mongodb          .zsh_history    Downloads
.config       .mysql_history    .zsh_sessions   dsa 3 odd sem
.coolpot      .nom              abc              Library
root@Vishnu:~#
```

4. cd

```
root@Vishnu:~# pwd
/Users/amrutha
root@Vishnu:~# cd ~/Documents
root@Vishnu:~/Documents# pwd
/Users/amrutha/Documents
```

5. cd ..

```
root@Vishnu:~# cd ..
root@Vishnu:~# pwd
/Users/amrutha
root@Vishnu:~#
```

6. mkdir

```
root@Vishnu:~# mkdir testfolder
root@Vishnu:~# ~ % ls
abc          bookflow_api      Documents        Music          Public
abc.c       bookflow_fastapi  Downloads       mysql          PyCharmMiscProject
abc.c#y     CREATE            dsa 3 odd sem   OSSP           PyCharmProjects
Applications data              Library         package-lock.json show
blockhack hackathon Desktop          Movies          Pictures        testfolder
```

7. rmdir

```
root@Vishnu:~# ~ % rmdir testfolder
root@Vishnu:~# ~ % ls
abc          bookflow_api      Documents        Music          Public
abc.c       bookflow_fastapi  Downloads       mysql          PyCharmMiscProject
abc.c#y     CREATE            dsa 3 odd sem   OSSP           PyCharmProjects
Applications data              Library         package-lock.json show
blockhack hackathon Desktop          Movies          Pictures
root@Vishnu:~# ~ %
```

8. find

```
root@Vishnu:~# mkdir -p ~/os_test
root@Vishnu:~# touch ~/os_test/sample.txt
root@Vishnu:~# find ~ -name "sample.txt"
find: /Users/amrutha/Music/Music: Operation not permitted
find: /Users/amrutha/Pictures/Photos Library.photoslibrary: Operation not permitted
/Users/amrutha/Desktop/TextHack/sample.txt
find: /Users/amrutha/Library/Application Support/CallHistoryTransactions: Operation not permitted
find: /Users/amrutha/Library/Application Support/CloudDocs: Operation not permitted
find: /Users/amrutha/Library/Application Support/com.apple.sharedfilelist: Operation not permitted
find: /Users/amrutha/Library/Application Support/Knowledge: Operation not permitted
find: /Users/amrutha/Library/Application Support/com.apple.TCC: Operation not permitted
find: /Users/amrutha/Library/Application Support/FileProvider: Operation not permitted
find: /Users/amrutha/Library/Application Support/AddressBook: Operation not permitted
find: /Users/amrutha/Library/Application Support/FaceTime: Operation not permitted
find: /Users/amrutha/Library/Application Support/DifferentialPrivacy: Operation not permitted
find: /Users/amrutha/Library/Application Support/com.apple.avfoundation/Frecents: Operation not permitted
find: /Users/amrutha/Library/Application Support/CallHistoryDB: Operation not permitted
find: /Users/amrutha/Library/Assistant/SiriVocabulary: Operation not permitted
find: /Users/amrutha/Library/Daemon Containers: Operation not permitted
find: /Users/amrutha/Library/Autosave Information: Operation not permitted
```

9. locate

```
amrutha@Am root@Vishnu:~#r ~ % locate sample.txt

WARNING: The locate database (/var/db/locate.database) does not exist.
To create the database, run the following command:

    sudo launchctl load -w /System/Library/LaunchDaemons/com.apple.locate.plist

Please be aware that the database can take some time to generate; once
the database has been created, this message will no longer appear.

amrutha@Am root@Vishnu:~# ~ % mdfind "sample.txt"
2026-08-12 18:52:35.453 mdfind[50593:9947903] [UserQueryParser] Loading keywords and predicates for locale "en_US"
2026-08-12 18:52:35.453 mdfind[50593:9947903] [UserQueryParser] Loading keywords and predicates for locale "en"
/Users/amrutha/os_test/sample.txt
```

TOPIC 3 : File Management

1. touch

```
root@Vishnu:~# touch
usage: touch [-A [-] [[hh]mm]SS] [-achm] [-r file] [-t [[CC]YY]MMDDhhmm[.SS]]
        [-d YYYY-MM-DDThh:mm:SS[.frac][tz]] file ...
root@Vishnu:~# ~ % touch test.txt
root@Vishnu:~# ls
abc                bookflow_fastapi    dsa 3 odd sem      OSSP                show
abc.c              CREATE              Library             package-lock.json   test.txt
abc.c#y            data                Movies              Pictures
Applications        Desktop              Music                Public
blockhack hackathon Documents            mysql                PyCharmMiscProject
bookflow_api        Downloads            os_test              PycharmProjects
root@Vishnu:~#
```

2. cat

```
root@Vishnu:~# echo "Hello from my Mac" > test.txt
root@Vishnu:~# cat test.txt
Hello from my Mac
root@Vishnu:~# cp test.txt copy.txt
root@Vishnu:~# ls
abc                bookflow_fastapi    Downloads            os_test              PycharmProjects
abc.c              copy.txt             dsa 3 odd sem        OSSP                show
abc.c#y            CREATE              Library              package-lock.json   test.txt
Applications        data                Movies                Pictures
blockhack hackathon Desktop              Music                Public
bookflow api        Documents            mysql                PyCharmMiscProject
root@Vishnu:~#
```

3. cp

```
root@Vishnu:~# cp test.txt copy.txt
root@Vishnu:~# ls
abc          bookflow_fastapi  Downloads        os_test        PycharmProjects
abc.c        copy.txt          dsa 3 odd sem   OSSP          show
abc.c#y     CREATE           Library          package-lock.json  test.txt
Applications data             Movies
blockhack hackathon Desktop         Music
bookflow api  Documents      mysql
root@Vishnu:~#
```

4. mv

```
root@Vishnu:~# mv copy.txt renamed.txt
root@Vishnu:~# ls
abc          bookflow_fastapi  dsa 3 odd sem   OSSP          renamed.txt
abc.c        CREATE           Library          package-lock.json  show
abc.c#y     data            Movies          Pictures       test.txt
Applications Desktop         Music
blockhack hackathon Documents      mysql
bookflow api  Downloads      os_test
root@Vishnu:~#
```

5. rm

```
root@Vishnu:~# rm renamed.txt
root@Vishnu:~# ls
abc          bookflow_fastapi  dsa 3 odd sem   OSSP          show
abc.c        CREATE           Library          package-lock.json  test.txt
abc.c#y     data            Movies          Pictures
Applications Desktop         Music
blockhack hackathon Documents      mysql
bookflow api  Downloads      os_test
root@Vishnu:~#
```

6. head

```
root@Vishnu:~# head test.txt
Line 1: Apple
Line 2: Banana
Line 3: Mango
Line 4: Orange
Line 5: Grapes
Line 6: Watermelon
Line 7: Pineapple
Line 8: Papaya
Line 9: Strawberry
Line 10: Guava

root@Vishnu:~# head -5 test.txt
Line 1: Apple
Line 2: Banana
Line 3: Mango
Line 4: Orange
Line 5: Grapes
root@Vishnu:~#
```


7.tail

```
root@Vishnu:~# tail test.txt
Line 1: Apple
Line 2: Banana
Line 3: Mango
Line 4: Orange
Line 5: Grapes
Line 6: Watermelon
Line 7: Pineapple
Line 8: Papaya
Line 9: Strawberry
Line 10: Guava
root@Vishnu:~# tail -5 test.txt
Line 6: Watermelon
Line 7: Pineapple
Line 8: Papaya
Line 9: Strawberry
Line 10: Guava
root@Vishnu:~#
```

8.more

```
root@Vishnu:~# more test.txt
Line 1: Apple
Line 2: Banana
Line 3: Mango
Line 4: Orange
Line 5: Grapes
Line 6: Watermelon
Line 7: Pineapple
Line 8: Papaya
Line 9: Strawberry
Line 10: Guava
```

9.less

```
Line 1: Apple
Line 2: Banana
Line 3: Mango
Line 4: Orange
Line 5: Grapes
Line 6: Watermelon
Line 7: Pineapple
Line 8: Papaya
Line 9: Strawberry
Line 10: Guava
```

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~

~

~

(END)

10. wc

Last login: Wed Aug 12 19:23:32 on ttys007

```
root@Vishnu:~# wc test.txt
```

```
10      30     159 test.txt
```

```
root@Vishnu:~# wc -l test.txt
```

```
10 test.txt
```

```
root@Vishnu:~# wc -w test.txt
```

```
30 test.txt
```

```
root@Vishnu:~# wc -c test.txt
```

```
159 test.txt
```

```
root@Vishnu:~#
```

11. diff

```
root@Vishnu:~# diff test.txt test2.txt
root@Vishnu:~# echo "Line 11: Kiwi" >> test2.txt
root@Vishnu:~# diff test.txt test2.txt
10a11
> Line 11: Kiwi
root@Vishnu:~#
```

12. sort

```
root@Vishnu:~# echo "Mango"
Mango
Apple
Banana
Apple
Orange
[Banana" > names.txt
root@Vishnu:~# cat names.txt
Mango
Apple
Banana
Apple
Orange
Banana
root@Vishnu:~# sort names.txt
Apple
Apple
Banana
Banana
Mango
Orange
root@Vishnu:~#
```

13. uniq

```
root@Vishnu:~# sort names.txt | uniq
Apple
Banana
Mango
Orange
```

14. grep

```
root@Vishnu:~# cat names.txt
Mango
Apple
Banana
Apple
Orange
Banana
root@Vishnu:~# grep "Apple" names.txt
Apple
Apple
root@Vishnu:~#
```

TOPIC 4 : File Editing

1. nano

```
root@Vishnu:~# nano editor_test.txt
root@Vishnu:~# cat editor_test.txt
Hello, this file was created using Nano.
root@Vishnu:~#
```

2. vi

```
Last login: Wed Aug 12 19:26:46 on ttys007
root@Vishnu:~# vi editor_test.txt
root@Vishnu:~# cat editor_test.txt
Hello, this file was created using Nano.
root@Vishnu:~#
```

TOPIC 5 : C Program Compilation and Execution

1. gcc

```
root@Vishnu:~# nano hello.c
root@Vishnu:~# gcc hello.c -o hello
root@Vishnu:~#
```


2. ./program

```
root@Vishnu:~# ./hello
Hello from C program!
root@Vishnu:~#
```

3. which

```
root@Vishnu:~# which gcc
/usr/bin/gcc
```

4. whereis

```
root@Vishnu:~# whereis
usage: whereis [-abmqu] [-BM dir ... -f] program ...
root@Vishnu:~# whereis gcc
gcc: /usr/bin/gcc
root@Vishnu:~# which
```

5. man

```
CLANG(1)                                Clang                                CLANG(1)

NAME
  clang - the Clang C, C++, and Objective-C compiler

SYNOPSIS
  clang [options] filename ...

DESCRIPTION
  clang is a C, C++, and Objective-C compiler which encompasses preprocessing, parsing, optimization, code generation, assembly, and linking. Depending on which high-level mode setting is passed, Clang will stop before doing a full link. While Clang is highly integrated, it is important to understand the stages of compilation, to understand how to invoke it. These stages are:

  Driver The clang executable is actually a small driver which controls the overall execution of other tools such as the compiler, assembler and linker. Typically you do not need to interact with the driver, but you transparently use it to run the other tools.

  Preprocessing
  This stage handles tokenization of the input source file, macro expansion, #include expansion and handling of other preprocessor directives. The output of this stage is typically called a ".i" (for C), ".ii" (for C++), ".mi" (for Objective-C), or ".mii" (for Objective-C++) file.

  Parsing and Semantic Analysis
  This stage parses the input file, translating preprocessor tokens into a parse tree. Once in the form of a parse tree, it applies semantic analysis to compute types for expressions as well and determine whether the code is well formed. This stage is responsible for generating most of the compiler warnings as well as parse errors. The output of this stage is an "Abstract Syntax Tree" (AST).
```

TOPIC 6 : Process Management

1. ps

```
root@Vishnu:~# ps
PID TTY          TIME CMD
60296 ttys007    0:00.02 -zsh
root@Vishnu:~#
```

2. ps -ef

```
root@Vishnu:~# ~ % ps -ef
      PID TTY          TIME CMD
      0   1   0   0 16Jul26 ??      77:09.02 /sbin/launchd
      0  511   1   0 16Jul26 ??      35:04.19 /usr/libexec/logd
      0  513   1   0 16Jul26 ??      1:52.77 /usr/libexec/UserEventAgent (System)
      0  515   1   0 16Jul26 ??      23:44.05 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks
/FSEvents.framework/Versions/A/Support/fsevents
      0  516   1   0 16Jul26 ??      1:30.71 /System/Library/PrivateFrameworks/MediaRemote.framework/Support/mediare
moted
      0  519   1   0 16Jul26 ??      0:45.03 /usr/sbin/systemstats --daemon
      0  523   1   0 16Jul26 ??      7:08.94 /usr/libexec/configd
      0  525   1   0 16Jul26 ??      5:01.31 /System/Library/CoreServices/powerd.bundle/powerd
      0  526   1   0 16Jul26 ??      0:00.03 /usr/libexec/IOMFB_bics_daemon
      0  530   1   0 16Jul26 ??      0:00.39 /usr/libexec/remoted
      0  534   1   0 16Jul26 ??      0:00.03 /System/Library/PrivateFrameworks/CoreSpeech.framework/corespeechd_syst
em
      0  536   1   0 16Jul26 ??      0:13.09 /usr/libexec/watchdogd
      0  540   1   0 16Jul26 ??      28:57.18 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks
/Metadata.framework/Versions/A/Support/mds
      0  542   1   0 16Jul26 ??      0:01.33 /usr/libexec/kernelmanagerd
      0  543   1   0 16Jul26 ??      4:39.14 /usr/libexec/diskarbitrationd
      0  546   1   0 16Jul26 ??      4:50.82 /usr/libexec/coreduetd
      0  547   1   0 16Jul26 ??      0:52.38 /usr/sbin/syslogd
      0  549   1   0 16Jul26 ??      0:53.66 /usr/libexec/thermalmonitord
      0  550   1   0 16Jul26 ??      4:25.35 /System/Library/PrivateFrameworks/CoreDuetContext.framework/Resources/c
ontextstored
      0  551   1   0 16Jul26 ??      4:00.00 /usr/libexec/xprotectd
      0  552   1   0 16Jul26 ??      27:35.38 /usr/libexec/opendirectoryd
      0  553   1   0 16Jul26 ??      2:33.68 /System/Library/PrivateFrameworks/ApplePushService.framework/apsd
      0  554   1   0 16Jul26 ??      58:23.08 /System/Library/CoreServices/launchservicesd
```

3. top

```
Processes: 819 total, 3 running, 816 sleeping, 4404 threads
Load Avg: 2.66, 2.97, 3.12 CPU usage: 16.62% user, 6.14% sys, 77.23% idle
SharedLibs: 642M resident, 127M data, 93M linkedit. MemRegions: 0 total, 0B resident, 0B private, 1646M shared.
PhysMem: 15G used (2476M wired, 4767M compressor), 107M unused.
VM: 370T vsize, 6144M framework vsize, 4471(0) swapins, 75960(0) swapouts.
Networks: packets: 211970048/930 in, 21412992/15G out. Disks: 35201957/790G read, 20322694/364G written.
19:59:21
PID    COMMAND     %CPU    TIME    #TH    #WQ    #PORTS MEM    PURG    CMPRS    PGRP    PPID    STATE    BOOSTS    %CPU_ME
587    WindowServer 51.4    18:11:48 24     5     10396+ 710M- 16M-    252M    587     1     sleeping *0[1]    1.43899
48745   Google Chrom 35.8    29:18.03 42/1    1     561-   599M+ 0B      250M-    96762   96762   running  *0[7]    0.00000
32277   promceefplug 24.4    10:45:01 14      5     252    228M  0B      49M     32270   32270   sleeping *1[5]    0.08337
32270   wpsoffice    20.6    09:19:09 82      1     1741+  1327M 112K    896M    32270   1     sleeping *1[16416] 11.7587
85893   promceefplug 14.0    31:46.04 22      1     253    47M+  0B      21M     32270   32270   sleeping *0[1]    0.00000
593     coreaudiod   13.5    85:09.04 14      2     8670   40M   0B      25M     593     1     sleeping *0[1]    0.15385
0       kernel_task  11.0    13:28:37 622/11 0      0      15M   0B      0B      0       0     running  *0[0]    0.00000
96893   Google Chrom 9.8     19:00.02 19      1     271    28M   0B      18M     96762   96762   sleeping *0[13701] 0.03369
61795   top          9.2     00:01.95 1/1     0      30     10M   0B      0B      61795   60296   running  *0[1]    0.00000
96803   Google Chrom 7.1     01:54:49 25      3     502    608M+ 32M-    136M    96762   96762   sleeping *1[31603] 0.76359
96762   Google Chrom 5.3     59:56.32 46      2     1851   381M+ 0B      184M-    96762   1     sleeping *4[4295] 0.04954
87577   Google Chrom 4.0     15:36.65 34      1     441    340M- 0B      165M-    96762   96762   sleeping *0[81] 0.00000
41981   wpsbackend   2.1     34:25.78 29      1     243    48M   0B      37M     41981   32270   sleeping *0[3468] 0.00000
81723   WhatsApp     2.1     03:01:39 36      5     3065-  1269M- 1440K   792M-    81723   1     sleeping *1846[31] 0.00000
470     con.apple.D 1.0     01:55:00 7       5     1105-  32M   0B      1734K    470     1     sleeping *0[1]    0.00000
```

4. kill

```
root@Vishnu:~# sleep 1000 &
[2] 62597
[3] 62757
zsh: no matches found: [2]
root@Vishnu:~# ps -p 62597
  PID TTY          TIME CMD
 62597 ttys007    0:00.00 sleep 1000
root@Vishnu:~# kill 62597
root@Vishnu:~#
[2] - terminated  sleep 1000
root@Vishnu:~# ps -p 62597
  PID TTY          TIME CMD
```

5. kill -9

```
root@Vishnu:~# sleep 1000 &
[2] 63098
root@Vishnu:~# ps -p 63025
  PID TTY          TIME CMD
root@Vishnu:~#
root@Vishnu:~# kill -9 63025
kill: kill 63025 failed: no such process
root@Vishnu:~# ps -p 63025
  PID TTY          TIME CMD
root@Vishnu:~#
```

Topic 7: Job Control

1. jobs

```
root@Vishnu:~# jobs
[1]  running  sleep 1000
[2]  running  sleep 1000
[3]  - running  sleep 1000
[4]  + running  sleep 1000
root@Vishnu:~#
```

2. bg

```
root@Vishnu:~# sleep 1000
^Z
zsh: suspended sleep 1000
root@Vishnu:~# bg
[5] - continued sleep 1000
root@Vishnu:~# jobs
[1]    running    sleep 1000
[2]    running    sleep 1000
[3]    running    sleep 1000
[4] + running    sleep 1000
[5] - running    sleep 1000
root@Vishnu:~#
```

3. fg

```
root@Vishnu:~# fg
[4] - running    sleep 1000
^C
root@Vishnu:~#
```

4. &

```
root@Vishnu:~# sleep 1000 &
[1] 62430
```

Topic 8: File Permissions and Links

1. chmod

```
root@Vishnu:~# touch permission.txt
root@Vishnu:~# ls -l permission.txt
-rw-r--r-- 1 amrutha staff 0 Aug 12 20:15 permission.txt
root@Vishnu:~# chmod 644 permission.txt
root@Vishnu:~# ls -l permission.txt
-rw-r--r-- 1 amrutha staff 0 Aug 12 20:15 permission.txt
root@Vishnu:~#
```

2. chmod 755

```
root@Vishnu:~# chmod 755 permission.txt
root@Vishnu:~# ls -l permission.txt
-rwxr-xr-x 1 amrutha staff 0 Aug 12 20:15 permission.txt
root@Vishnu:~#
```

3. ln

```
root@Vishnu:~# echo "Hello from original file" > original.txt
root@Vishnu:~#
[1] done sleep 1000
root@Vishnu:~# ln original.txt hardlink.txt
root@Vishnu:~# ls -l original.txt hardlink.txt
-rw-r--r-- 2 amrutha staff 25 Aug 12 20:19 hardlink.txt
-rw-r--r-- 2 amrutha staff 25 Aug 12 20:19 original.txt
root@Vishnu:~# cat hardlink.txt
Hello from original file
root@Vishnu:~#
```

4. ln -s

```
root@Vishnu:~# ln -s original.txt symlink.txt
root@Vishnu:~# ls -l symlink.txt
lrwxr-xr-x 1 amrutha staff 12 Aug 12 20:20 symlink.txt -> original.txt
root@Vishnu:~# cat symlink.txt
Hello from original file
root@Vishnu:~#
```

Topic 9: Disk and Memory Information

1. df -h

```
root@Vishnu:~# df -h
Filesystem      Size  Used Avail Capacity iused ifree %iused  Mounted on
/dev/disk3s1s1  460Gi  12Gi  318Gi    4%   459k  3.3G    0%    /
devfs           213Ki  213Ki   0Bi  100%    736    0  100%    /dev
/dev/disk3s6    460Gi  2.0Gi  318Gi    1%     2  3.3G    0%    /System/Volumes/VM
/dev/disk3s2    460Gi  8.4Gi  318Gi    3%   1.4k  3.3G    0%    /System/Volumes/Preboot
/dev/disk3s4    460Gi  2.4Mi  318Gi    1%    65  3.3G    0%    /System/Volumes/Update
/dev/disk1s2    500Mi  6.0Mi  482Mi    2%     1  4.9M    0%    /System/Volumes/xarts
/dev/disk1s1    500Mi  6.0Mi  482Mi    2%    33  4.9M    0%    /System/Volumes/iSCPreboot
/dev/disk1s3    500Mi  1.4Mi  482Mi    1%    94  4.9M    0%    /System/Volumes/Hardware
/dev/disk3s5    460Gi  119Gi  318Gi   28%   1.3M  3.3G    0%    /System/Volumes/Data
map auto_home   0Bi    0Bi   0Bi  100%     0    0    -    /System/Volumes/Data/home
/dev/disk5s1    677Mi  525Mi  148Mi   78%   7.9k  1.5M    1%    /Volumes/Arduino IDE 2.3.10-arm64
/dev/disk6s1    1.3Gi  1.3Gi  2.4Mi  100%   11k  4.3G    0%    /Volumes/ChatGPT Installer
root@Vishnu:~#
```


2. du -sh

```
root@Vishnu:~# du -sh .
du: ./Music/Music: Operation not permitted
du: ./Pictures/Photos Library.photoslibrary: Operation not permitted
du: ./Library/Application Support/CallHistoryTransactions: Operation not permitted
du: ./Library/Application Support/CloudDocs: Operation not permitted
du: ./Library/Application Support/com.apple.sharedfilelist: Operation not permitted
du: ./Library/Application Support/Knowledge: Operation not permitted
du: ./Library/Application Support/com.apple.TCC: Operation not permitted
du: ./Library/Application Support/FileProvider: Operation not permitted
du: ./Library/Application Support/AddressBook: Operation not permitted
du: ./Library/Application Support/FaceTime: Operation not permitted
du: ./Library/Application Support/DifferentialPrivacy: Operation not permitted
du: ./Library/Application Support/com.apple.avfoundation/Frecents: Operation not permitted
du: ./Library/Application Support/CallHistoryDB: Operation not permitted
du: ./Library/Assistant/SiriVocabulary: Operation not permitted
du: ./Library/Daemon Containers: Operation not permitted
du: ./Library/Autosave Information: Operation not permitted
du: ./Library/IdentityServices: Operation not permitted
du: ./Library/Messages: Operation not permitted
du: ./Library/HomeKit: Operation not permitted
du: ./Library/Sharing: Operation not permitted
du: ./Library/com.apple.aiml.instrumentation: Operation not permitted
du: ./Library/Mail: Operation not permitted
du: ./Library/Trial: Operation not permitted
du: ./Library/AppleMediaServices: Operation not permitted
du: ./Library/DuetExpertCenter: Operation not permitted
du: ./Library/Accounts: Operation not permitted
du: ./Library/Safari: Operation not permitted
du: ./Library/Bioime: Operation not permitted
du: ./Library/IntelligencePlatform: Operation not permitted
```

3. free -h

```
root@Vishnu:~# top -l 1 | grep PhysMem
PhysMem: 15G used (2709M wired, 5195M compressor), 177M unused.
root@Vishnu:~#
```

Topic 10: User and Environment Management

1.id

```
root@Vishnu:~# id
uid=501(amrutha) gid=20(staff) groups=20(staff),12(everyone),61(localaccounts),79(_appserverusr),80(admin),81(_appserver
adm),33(_appstore),98(_lpadmin),100(_lpoperator),204(_developer),250(_analyticsusers),395(com.apple.access_ftp),398(com.
apple.access_screensharing),399(com.apple.access_ssh),400(com.apple.access_remote_ae),701(com.apple.sharepoint.group.1)
root@Vishnu:~#
```

2.env


```

amrutha@Amruthas-MacBook-Air ~ % env
TMPDIR=/var/folders/h1/gvz4hvx5wn7l5_tmqw_qv8r0000gn/T/
CFBundleIdentifier=com.apple.Terminal
XPC_FLAGS=0x0
TERM=xterm-256color
SSH_AUTH_SOCK=/var/run/com.apple.launchd.WSLs4dcfr/Listeners
XPC_SERVICE_NAME=0
TERM_PROGRAM=Apple_Terminal
COLORTERM=truecolor
OSLogDataLimit=64
TERM_PROGRAM_VERSION=470.2
TERM_SESSION_ID=862A7243-698C-4EB3-9984-AF70AC18D51E
SHELL=/bin/zsh
HOME=/Users/amrutha
LOGNAME=amrutha
USER=amrutha
PATH=/opt/homebrew/opt/postgresql17/bin:/opt/homebrew/bin:/opt/homebrew/sbin:/Library/Frameworks/Python.framework/Versions/3.14/bin:/usr/local/bin:/System/Cryptexes/App/usr/bin:/usr/bin:/bin:/usr/sbin:/sbin:/var/run/com.apple.security.cryptextd/codex.system/bootstrap/usr/local/bin:/var/run/com.apple.security.cryptextd/codex.system/bootstrap/usr/bin:/var/run/com.apple.security.cryptextd/codex.system/bootstrap/usr/appleinternal/bin:/pkg/env/global/bin
SHELVL=1
PWD=/Users/amrutha
OLDPWD=/Users/amrutha
HOMEBREW_PREFIX=/opt/homebrew
HOMEBREW_CELLAR=/opt/homebrew/Cellar
HOMEBREW_REPOSITORY=/opt/homebrew
FPATH=/opt/homebrew/share/zsh/site-functions:/usr/local/share/zsh/site-functions:/usr/share/zsh/site-functions:/usr/share/zsh/5.9/functions
INFOPath=/opt/homebrew/share/info:
LANG=en_US.UTF-8
_/usr/bin/env
amrutha@Amruthas-MacBook-Air ~ %

```

3. export

```

root@Vishnu:~# export MYNAME="Amrutha"
root@Vishnu:~# echo $MYNAME
Amrutha
root@Vishnu:~#

```

4. alias

```

root@Vishnu:~# alias ll='ls -l'
root@Vishnu:~# ll
total 272
-rwxr-xr-x 1 amrutha staff 33424 Jul 31 13:54 abc
-rw-r--r-- 1 amrutha staff 80 Jul 31 13:51 abc.c
-rw-r--r-- 1 amrutha staff 81 Jul 29 12:39 abc.c#y
drwx-----@ 5 amrutha staff 160 Jun 19 10:23 Applications
drwxr-xr-x 2 amrutha staff 64 Aug 7 09:39 blockhack hackathon
drwxr-xr-x 3 amrutha staff 96 Aug 12 15:31 bookflow_api
drwxr-xr-x 5 amrutha staff 160 Aug 12 17:20 bookflow_fastapi
-rw-r--r-- 1 amrutha staff 17827 Jul 30 11:02 CREATE
drwxr-xr-x 3 amrutha staff 96 Aug 3 14:53 data
drwx-----@ 141 amrutha staff 4512 Aug 12 19:59 Desktop
drwx-----@ 5 amrutha staff 160 Aug 11 13:52 Documents
drwx-----@ 337 amrutha staff 10784 Aug 12 18:09 Downloads
drwxr-xr-x 3 amrutha staff 96 Jul 23 12:20 dsa 3 odd sem
-rw-r--r-- 1 amrutha staff 41 Aug 12 19:41 editor_test.txt
-rw-r--r-- 2 amrutha staff 25 Aug 12 20:19 hardlink.txt
-rwxr-xr-x 1 amrutha staff 33432 Aug 12 19:43 hello
-rw-r--r-- 1 amrutha staff 88 Aug 12 19:43 hello.c
drwx-----@ 91 amrutha staff 2912 Jul 25 08:50 Library
drwx----- 5 amrutha staff 160 Apr 22 21:03 Movies
drwx-----+ 4 amrutha staff 128 Mar 20 10:47 Music
-rw-r--r-- 1 amrutha staff 51 Jul 30 11:02 mysql
-rw-r--r-- 1 amrutha staff 39 Aug 12 19:29 names.txt
-rw-r--r-- 2 amrutha staff 25 Aug 12 20:19 original.txt
drwxr-xr-x 3 amrutha staff 96 Aug 12 18:50 os_test
drwxr-xr-x 17 amrutha staff 544 Aug 3 12:41 OSSP
-rw-r--r--@ 1 amrutha staff 86 May 11 10:41 package-lock.json
-rwxr-xr-x 1 amrutha staff 0 Aug 12 20:15 permission.txt
drwx-----+ 6 amrutha staff 192 Jun 13 20:31 Pictures
drwxr-xr-x+ 4 amrutha staff 128 Feb 21 12:34 Public
drwxr-xr-x@ 4 amrutha staff 128 Apr 16 14:18 PyCharmMiscProject
drwxr-xr-x@ 3 amrutha staff 96 Apr 16 14:18 PycharmProjects
-rw-r--r-- 1 amrutha staff 0 Jul 30 11:03 show
lrwxr-xr-x 1 amrutha staff 12 Aug 12 20:20 symlink.txt -> original.txt
-rw-r--r-- 1 amrutha staff 159 Aug 12 19:07 test.txt
-rw-r--r-- 1 amrutha staff 173 Aug 12 19:28 test2.txt
root@Vishnu:~#

```